

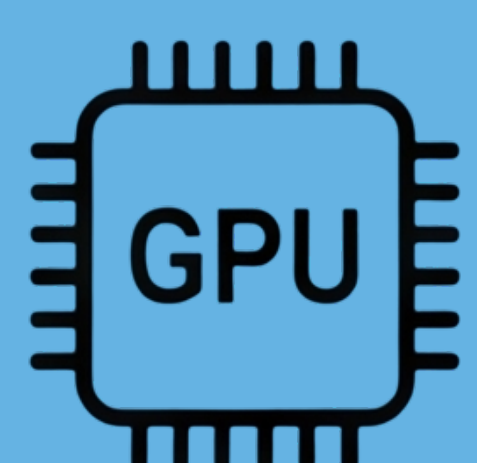
Prefill Phase

Input Block

System
Context

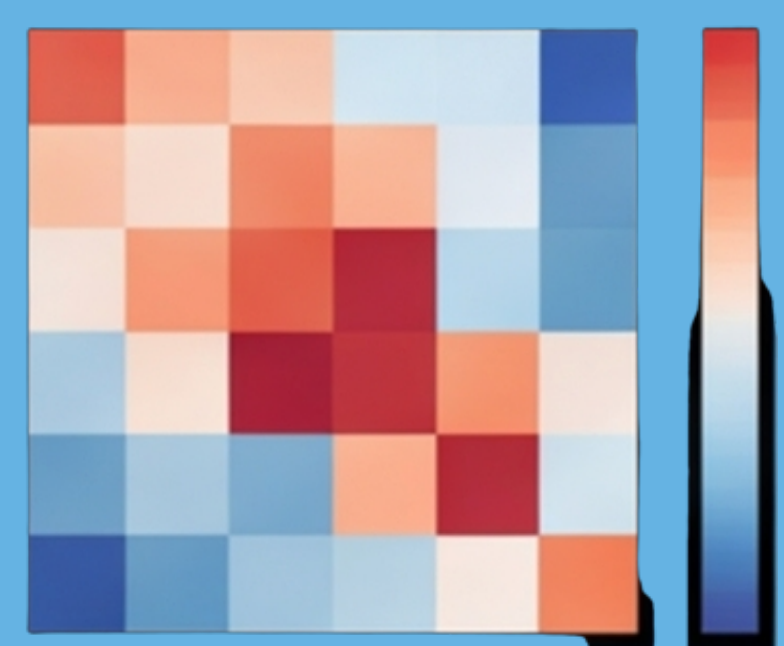
User
Context

User
Query



Full Forward Pass

Attention Tracker



Accumulate
attention scores

Semantic Analyzer

Info Density

Query Relevance

Factual Likelihood



Pinning Mask



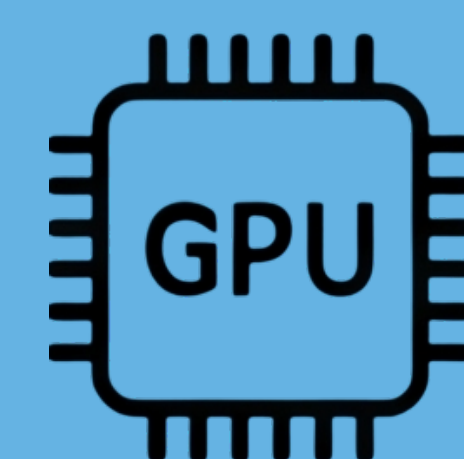
initial KV
Cache

static
signals

Attention Signals

Content Signals

Decode Phase



Single-Token Forward Pass

process token at a time

Attention Update

$$\sum_{\text{layer}}^n \sum_{j=1}^{l_j} \sum_{\text{head}}^m A_{ijl} t_{\square h}$$

$$\lambda = \frac{\text{head entropy}}{\log(\lambda - h)\sigma_i)}$$

no

Budget Check

yes

pruned
cache
state

Multi-Signal Scoring

$$S = \sum_{\text{layers}}^m \text{attention} + \sum_{n=h} w_{ig}(\text{content} + -\infty)$$

Pinned tokens = ∞

Block Selection and Eviction

Cache blocks are grouped, scored by max member & pruned to retain top-k blocks

Scoring & Eviction

Protected Tokens